

BENG 285/BNFO 285/ECE 204. Statistical Learning in Bioinformatics

Project 1: Practical understanding of approaches for dimension reduction of multi-dimensional datasets.

Learning objectives: The overall learning objective of this project is to utilize dimensionality reduction approaches to multi-dimensional datasets for identifying meaningful subgroups. The project will require for students to apply both linear and non-linear dimensionality reduction methods, including principal-component analysis (PCA), multi-dimensional scaling (MDS), t-distributed stochastic neighbor embedding (t-SNE), and uniform manifold approximation and projection (UMAP). Students will likely need to consider which approach is more appropriate for their particular dataset and what constitutes a meaningful subgroup. Thus, students will likely need to utilize nonparametric and/or parametric statistical tests to evaluate whether any of the approaches is separating samples based on the available patients' metadata. Lastly, students will need to compare the differences in the results and groupings yielded by each of the four methods.

Subject Matter: The focus of project 1 will be on dimensionality reduction of gene expression data derived from cancer samples. While there could be gene expression data available for as many as 25,000 protein coding genes in each cancer sample, examining many thousands of features for a cancer sample is impractical and, in many cases, impossible. To allow better understanding of the different subgroups present in each dataset, students will need to apply four dimensionality reduction approaches (namely, PCA, MDS, t-SNE, and UMAP) to the provided data. After applying each of these approaches, students will need to compare the results and explain the observed differences between the approaches. Importantly, students should try to evaluate the different subgroups yielded by each of the methods and perform comparisons amongst these subgroups. The comparisons should reveal whether there are any statistically significant differences in regards to patients' metadata (*e.g.*, cancer subtype, age of diagnosis, sex, race, clinical endpoint, center generating the data, *etc.*).

Project's manuscript: <https://www.sciencedirect.com/science/article/pii/S2211124721008597>

Manuscript presentation: Team presentation on Tuesday (8-Apr) at 6:30pm (CENTR 217B)

Project's datasets: <https://www.dropbox.com/scl/fo/7d37xqur5v1b8jni61b0t/AC1f0-cbGkGBSZFI9fXzdQ?rlkey=pfw7xmb7slnz7d398gzfzpu7>

Project's presentation: Team presentation on Thursday (17-April) at 6:30pm (CENTR 217B)

Deadline: All teams submit a project report (max 3 pages) by 11:59pm on Friday (18-April)

Potential outline for approaching the project:

- Apply each of the four dimensionality reduction approaches (PCA, MDS, t-SNE, and UMAP) to your data. Be careful of any parameters when applying the approaches.
- For each approach, where applicable, you will need to explore the optimal number of components. For example, why should one select to evaluate the two or three principal components instead of examining another number of principal components.
- Students should outline the differences between each of the methods as observed in their dataset and comment on which method provides more meaningful and/or useful results.
- One way to compare the methods is by evaluating whether there are significant differences in regards to the available patients' metadata.
- Students should also evaluate whether any groupings are likely artifactual, for example, due to batch effects from the centers generating these data.